

Data Engineer Take-Home Assignment

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10-05-2026

Part 2: Requirements Gathering and Problem Space Analysis

We need to understand what factors lead to successful career growth across different industries.

Clarifying Questions

1. How do you define successful career growth?

Is it reaching promotion faster, a large increase in salary, or something else? This question will help us understand the main measure for career growth. If it is speed, I would measure time spent in a job before moving up. If it is salary, the change in salary bands between jobs.

2. Are salary bands comparable across companies and industries?

Each job in the source data has a salary band from 1 to 6, but nothing explains what a band represents. Is band 3 in Finance equivalent to band 3 in Technology? Every job move in the data is exactly one band up, so the band alone cannot show who had the bigger increase. If bands are not equivalent, I would need the salary range behind each band to compare industries fairly.

3. Who will use the results, and in what form?

Are they summary numbers for business leaders, or will support agents use them to give recommendations to individual job seekers? The audience will help us decide what needs to be delivered and how often the data needs to be refreshed.

Key Metrics

My assumption about this question is what a job seeker would want to know about their own career

1. Salary band by years of experience, within an industry

This will tell a job seeker where they stand next to people with similar experience in the same industry. Compare years of experience so that someone 3 years is not measured against someone 18 years in.

2. Career growth by skill, education level and certifications

This metric shows which skills, degrees and certifications go along with faster growth, and these are things a job seeker can act on. In the sample, master's holders moved up quicker and held the most certifications. This is just a pattern in the data provided.

3. Time in role before promotion, by industry

This helps a job seeker understand how long a step up usually takes, so they can judge whether they are on track. In the sample it is about 27 months in Marketing and about 41 in Finance.

How the Data Model Supports the Analysis

I have designed a dimensional model with three fact tables and five dimensions.

Fact tables

1. **fact_job** has one row per job a person held. It holds the salary band, tenure, band change from the previous job and time spent in the previous job. It links to the professional, company, industry and role dimensions.
2. **fact_professional_skill** has one row per skill a person has, with their proficiency level. It links to the professional and skill dimensions.
3. **fact_certification** has one row per certification a person earned. It links to the professional dimension.

Dimensions

1. **dim_professional** has one row per person, with their education level, current role, current industry and certification count. It is shared by all three fact tables, which is what lets me combine a person's jobs with their skills and certifications.
2. **dim_company** has one row per company.
3. **dim_industry** has one row per industry.
4. **dim_role** has one row per job title.
5. **dim_skill** has one row per skill. It is keyed on the skill name, because the source gives the same skill different IDs.

Each metric comes straight from the model:

1. Salary band by years of experience uses the salary band and dates on **fact_job**, grouped by industry.
2. Growth by skill, education and certifications links a person's jobs to their skills and certifications through **dim_professional**.
3. Time in role before promotion uses days_in_previous_job on **fact_job**, grouped by industry.

Additional Data Sources

1. Salary ranges per band and industry

Every job move in the data is exactly one band up, so the band alone cannot show who had the bigger increase. Salary ranges would let me compare industries fairly.

2. A standard list of job titles and skills

I match roles and skills by exact name, and almost every job title in the sample is unique. A standard list would group them into common roles, so metrics by role become meaningful.

3. A last-updated date on each record

The file does not say when it was captured, so I had to assume a reference date for current jobs. A last-updated date would fix that and would also let the pipeline load only the records that changed.

Part 3: Airflow Integration & Production Considerations

Production Considerations

Things I would change to run this in production:

1. Load incrementally, each run picks up only the records updated since the last run, which needs a last-updated date on the source records.
2. Load the first run in batches, since the worker only has 2 GB of memory.
3. Replace a person's rows on each load (delete and insert), so a rerun gives the same result.
4. Use hash keys for the dimensions. The numbered keys in the notebook would shift when new companies or skills arrive.
5. Retry short outages, but stop and alert when the quality checks fail, because bad data needs manual intervention.
6. Use the run date as the reference date for current jobs.
7. Move the notebook logic into a Python module that the DAG tasks can call.